Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/04/2016
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2012	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	6
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 5/31/2017 per NSSC information (Ed., 6/17/13) NOTE: Change in grant number and to be extended, per D. Chao/GRC (Ed. 3/6/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AB01G ; NNX12AK14G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for International Space Station (ISS) experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	The fourth year of this project included a number of efforts addressing flow boiling, critical heat flux (CHF), and condensation in microgravity. This report summarizes key findings from the fourth year, which are presented in two parts. Part 1 addresses the development of a consolidated methodology to predicting flow boiling CHF for inclined channels in Earth gravity and for microgravity. It consists of a comprehensive investigation of the complex trends of flow boiling CHF in a rectangular channel in both gravitational environments. It is shown that the Interfacial Lift-off Model provides good predictions of CHF data for both single-sided and double-sided heating. CHF mechanism in Earth gravity is shown to be highly sensitive to flow orientation at very low velocities, but is consistent with the wavy vapor layer depiction of the Interfacial Lift-off Model at high velocities. The model predicts a stable vapor-liquid interface for downflow with a downward-facing heated wall at lower velocities, and wavy interface with a critical wavelength that decreases with increasing velocity at higher velocities. Predicted CHF values for microgravity fall about midway between the maxima and minima for Earth gravity. Overall, predicted values of CHF and key interfacial parameters for all orientations in Earth gravity and for microgravity converge around 1.5 m/s, which points to a velocity threshold above which inertia begins to effectively negate gravity effects. Part 2 reviews published literature concerning two-phase flow and heat transfer in reduced gravity. Discussed are the different methods and platforms dedicated to exploring the influence of reduced gravity, including ground flow boiling experiments performed at different orientations relative to Earth gravity, as well as reduced gravity adiabatic two-phase flow, pool boiling, flow boiling, and CHF experiments. Despite the extensive data and flow visualization results available in the literature, it is shown that there is a severe shortage of useful correlations, mechanistic models, and computational models, which compromises readiness to adopt flow boiling in future space systems. Key recommendations are provided concerning platform, heater design, and operating conditions for future studies to expedite the deployment of two-phase thermal management in future space missions.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Mascarenhas N, Lee H, Mudawar I. "Experimental and computational investigation of interfacial shear along a wavy two-phase interface." International Journal of Heat and Mass Transfer. 2015 Jun;85:265-80. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2015.01.096 , Jun-2015
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Articles in Peer-reviewed Journals	Kim SM, Mudawar I. "Review of two-phase critical flow models and investigation of the relationship between choking, premature CHF, and CHF in micro-channel heat sinks." International Journal of Heat and Mass Transfer. 2015 Aug;87:497-511. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2015.04.012 , Aug-2015
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Awards	Mudawar I. "Thomson Reuters Highly Cited Researcher, December 2015." Dec-2015

Awards	Mudawar I. "Betty Ruth and Milton B. Hollander Family Professor of Mechanical Engineering, December 2015." Dec-2015
Awards	Mudawar I. "Thomson Reuters list of “The World’s Most Influential Scientific Minds 2015”, December 2015." Dec-2015